

- Object Inode Number = 33001
 - Object Security Context = system_uobject_r:tmp_tz:0
3. Access details in the log file: Operation being Performed by Subject{ described in 1) above } on Object{ described in 2) above } = read
4. Result: Result of the Operation{ described in 3) above } = denied

The above can also be viewed in an easy-to-understand GUI. To be able to view the log file in GUI mode, install the **setools-gui** package.

Once installed, you can view your log file by executing the **seaudit** command. A GUI window shall open, which is similar to Figure 1.

In the file menu, select **Open Log** and then select **/var/log/audit/audit.log**.

Once you open the log file, you can see the details of error logs generated graphically, sorted in various columns. You can right click on the message of your choice and look at the original log message.

Resolving SELinux errors

Once you have decoded the error from the log file, SELinux provides a plethora of tools to help solve the errors. Depending on the solutions and the security level desired by a systems administrator, the possible solutions can be applied to allow the action. The **setroubleshoot** RPM will be required for this.

The **setroubleshoot** RPM is an important tool in the SELinux systems administrator's toolkit. If it is not installed in your system, please go ahead and install it. **sealert** is one of the utilities provided.

To understand **sealert** usage, issue the following command:

```
-S --no-service Start sealert without dbus service as stand alone app
-i --lookup-id Lookup alert by id
-a --analyze-file Scan a log file, analyze it's AVC's
-H --html-output Output in html
-v --verbose Start in verbose mode
```

Let us first analyse the log file **/var/log/audit/audit.log** using **sealert**. To do so, enter the command that follows:

```
[root@vbg selinux]# sealert -a /var/log/audit/audit.log
```

Depending on the number of errors in the log file, an output containing the possible solutions to AVC denial errors in the log file will appear on the screen. To analyse these solutions, you can redirect the output to a file:

```
[root@vbg selinux]# sealert -a /var/log/audit/audit.log > \
/tmp/tmp_selinux_error-solutions.txt
```

Each error and possible solution is mentioned in the text file (**/tmp/my-selinux-error-solutions.txt**) separated by a line of dashes. The AVC error example discussed in the above section has been analysed by **sealert** as:

Summary

SELinux is preventing the **/usr/sbin/httpd** from using potentially mislabeled files (**/usr/www/html/index.html**).

Detailed Description

SELinux has denied **/usr/sbin/httpd** access to potentially mislabeled file(s) (**/usr/www/html/index.html**). This means that SELinux will not allow **/usr/sbin/httpd** to use these files. It is common for users to edit files in their home directory or **/tmp** directories and then move (**mv**) them to system directories. The problem is that the files end up with the wrong file context which confined applications are not allowed to access.

Allowing Access

If you want **/usr/sbin/httpd** to access this files, you need to relabel them using **restorecon -v /usr/www/html/index.html**. You might want to relabel the entire directory using **restorecon -R -v /usr/www/html**.

Additional Information

```
Source Context      user_system_r:httpd_t:0
Target Context      system_uobject_r:tmp_t:0
Target Objects      /usr/www/html/index.html [file]
Affected RPM Packages
Policy RPM
SELinux Enabled
Policy Type
MLS Enabled
Enforcing Mode
Plugin Name         plugins/home_tmp_bad_labels
Host Name
Platform
Alert Count         2
Line Numbers        8796,8797,8798
```

31755,0-1 70%

```
[root@vbg selinux]# sealert -h
-b --browser Launch the browser
-h --help Show this message
-s --service Start sealert as a dbus service
```